

A Day in the Life of a SOC Analyst



Arif Hussain
Cybersecurity Professional
LinkedIn: [linkedin.com/in/arifhussainofficial](https://www.linkedin.com/in/arifhussainofficial)
Email: Arifhussain1@outlook.com

Why do we need to establish a SOC

Organizations today face constant cyber threats — including attacks, malware, phishing, ransomware, insider threats, and more. A Security Operations Center (SOC) provides a centralized team and system dedicated to detecting, analyzing, responding to, and recovering from these threats in real time.

Here are key reasons why a SOC is important:

- 24/7 Threat Monitoring
- Faster Incident Response
- Compliance
- Threat Intelligence
- Reduced Damage
- Continuous Improvement

In simple words:

Without a SOC = delayed detection, slower response, higher risk.

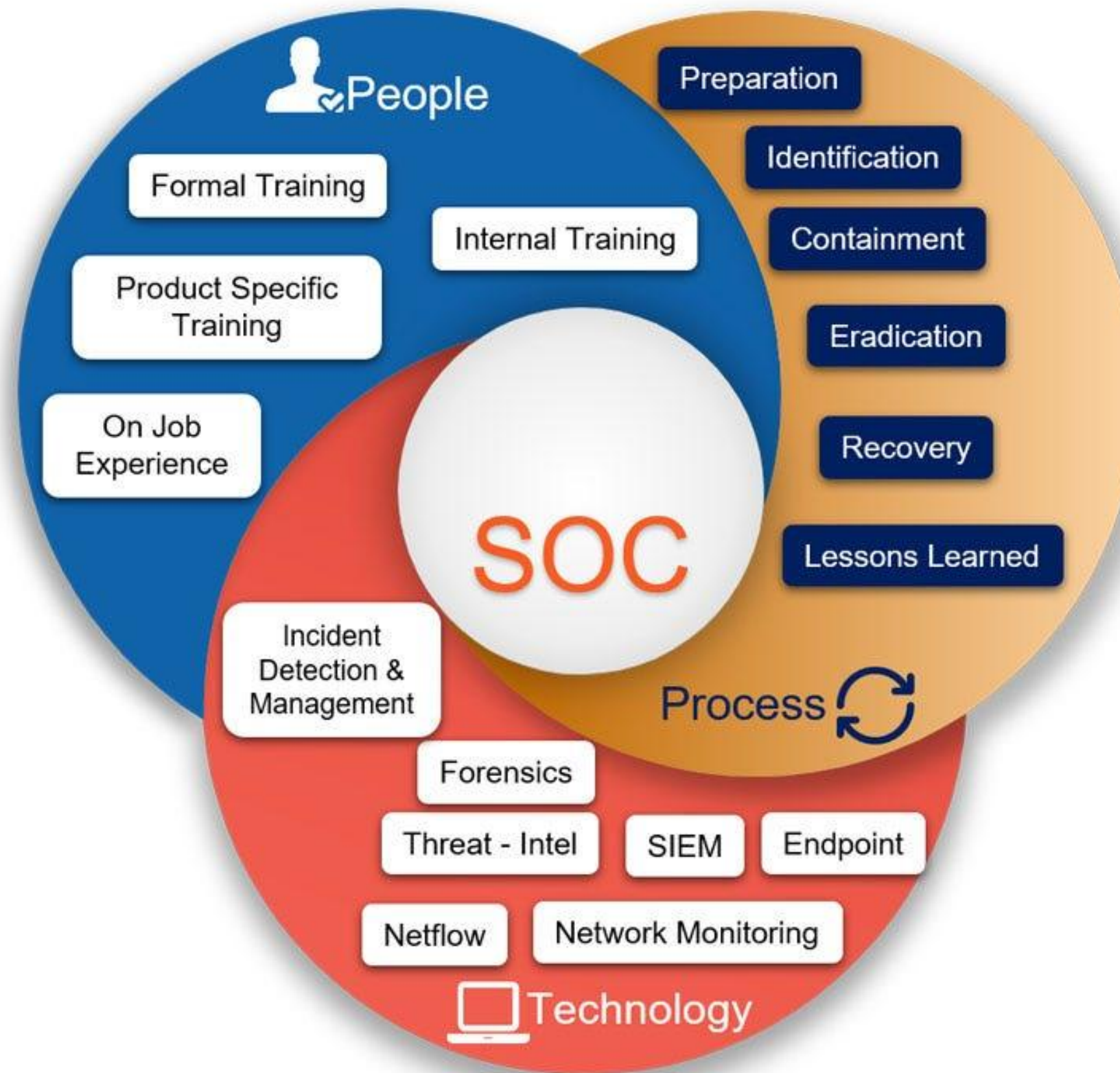
With a SOC = proactive defense, faster action, stronger security posture.

Compliance and Framework Requirements

Several frameworks and authorities require or recommend having a SOC:

- ISO/IEC 27001:2022
- NIST Cybersecurity Framework (CSF)
- PCI DSS
- HIPAA
- GDPR
- Basel II/III (for Financial Institutions)
- Local Regulators (e.g., SBP in Pakistan)

SOC Functions



11 Strategies of a World-Class SOC

- Know what you are protecting and why
- Give the SOC the authority to do its job
- Build a SOC structure to match your organizational needs
- Hire and grow quality staff
- Prioritize incident response
- Illuminate adversaries with cyber threat intelligence
- Select and collect the right data
- Leverage tools to support analyst workflow
- Communicate clearly, collaborate often, share generously
- Measure performance to improve performance
- Turn up the volume by expanding SOC functionality

Structure of a Modern SOC

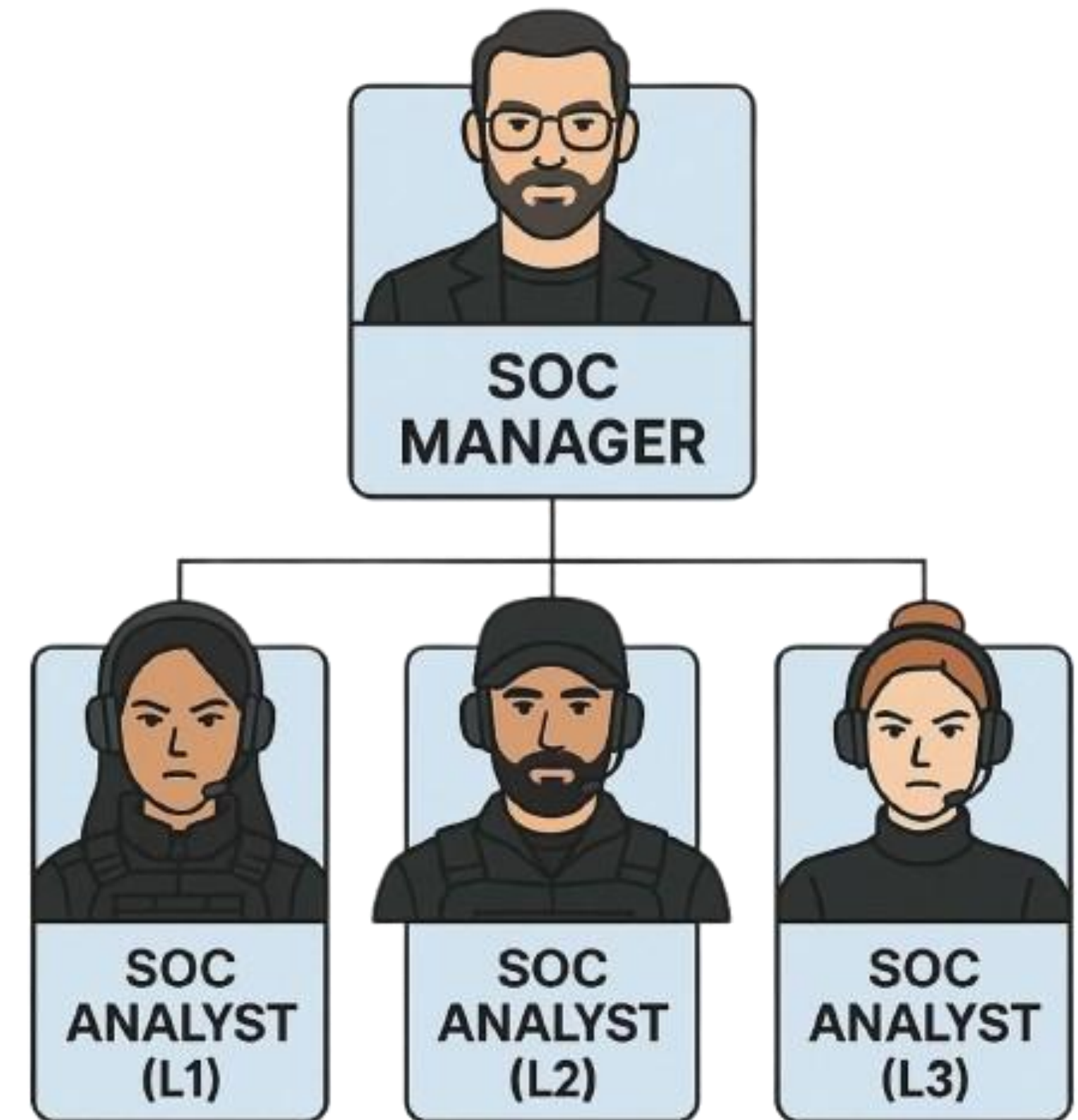
Level 1: Alert Monitoring & Triage

Level 2: Deep Incident Analysis

Level 3: Threat Hunting, Forensics

SOC Management:

- Governance & Metrics
- Fusion Center Integration
- Coordination with interested parties



Core Tools in the SOC Arsenal

SIEM: Loghrythm, Sentinel, QRadar

SOAR: XSOAR, IBM Resilient

EDR/XDR: CrowdStrike, Microsoft Defender XDR

TIPs: AlienVault, USCert

Cloud Security: GuardDuty, Azure Defender

DRP: Recorded Future, CTM360 etc

Critical Controls to Integrate

- MITRE ATT&CK Mapping
- Threat Intelligence Integration
- Purple Team Exercises
- Automatic Playbook Updates
- Data Privacy & Insider Threat Monitoring
- Deepfake Detection

The Future SOC: Intelligence-Driven and Adaptive

- Predictive Threat Modeling
- Adaptive Automated Response
- Complete Asset Visibility
- DevSecOps Integration
- AI Copilots for Analysts

Key Metrics to Track

- Mean Time to Detect (MTTD)
- Mean Time to Respond (MTTR)
- Incidents by severity
- SOC Analyst workload
- Detection fidelity (FP/TP ratio)
- Dwell time reduction

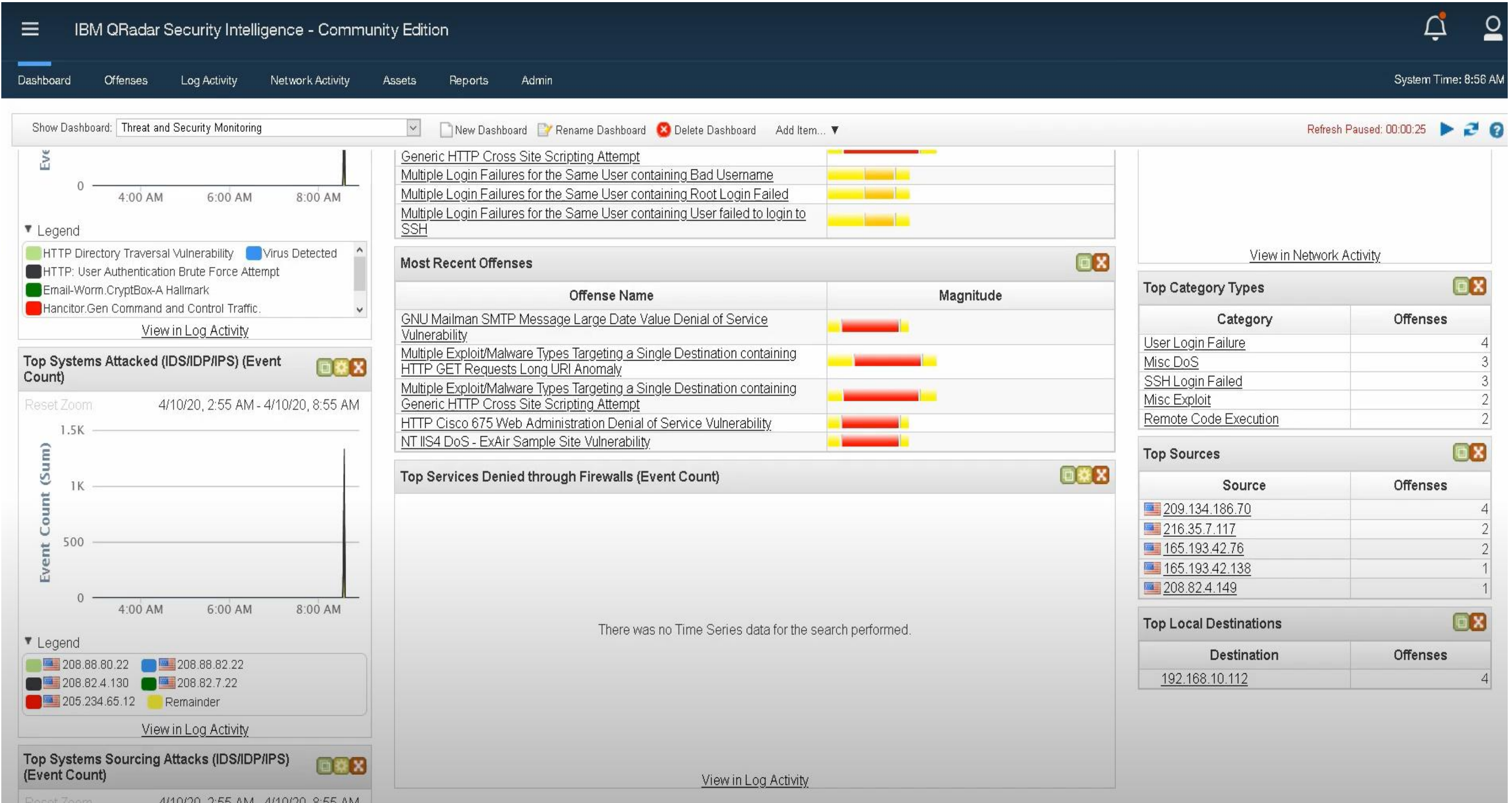
Challenges in the SOC

- Alert fatigue
- Skills shortage and burnout
- Sophisticated attacks
- Budget constraints
- Cloud visibility challenges

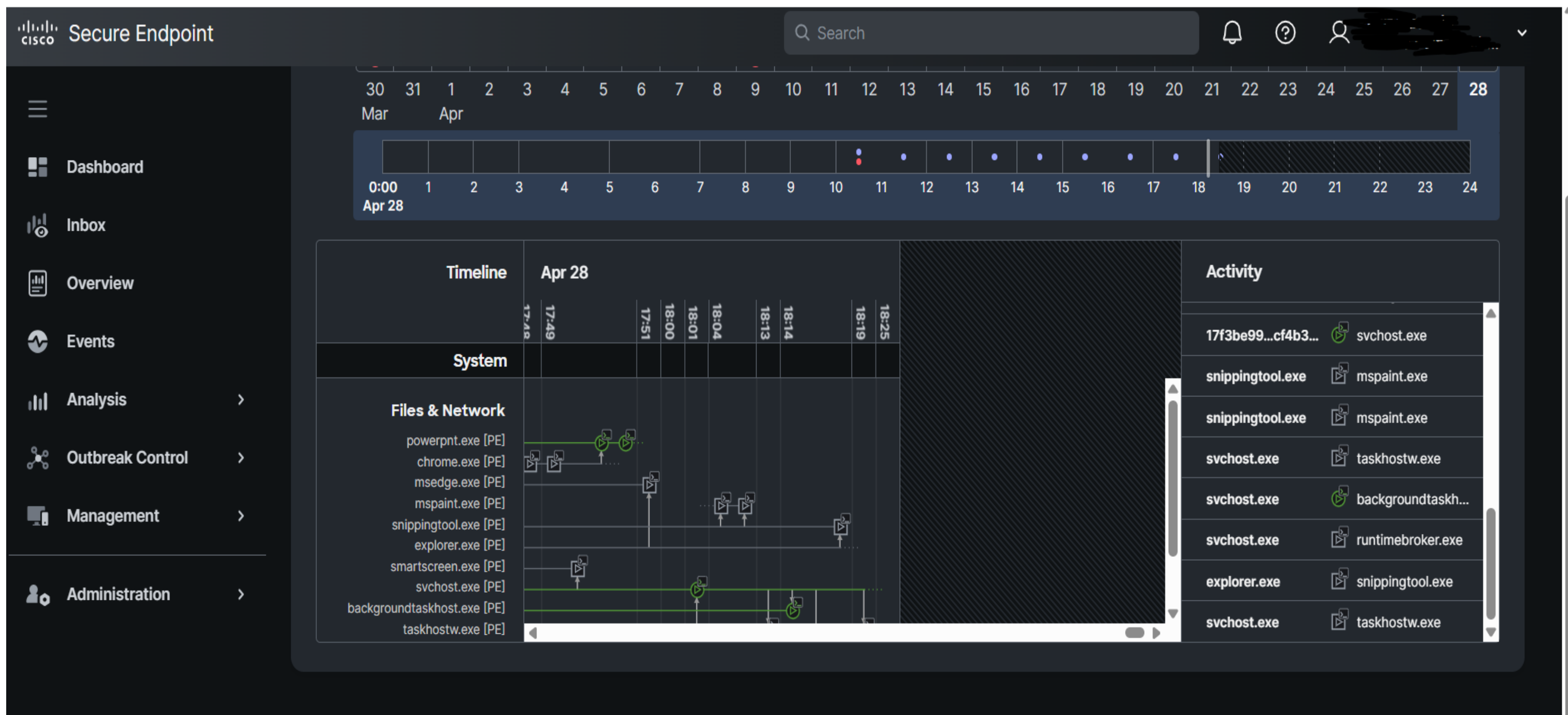
A Day in the Life of a SOC Analyst

- Shift Handover Briefing
- Log and Alert Review
- Triage
- Threat Hunting
- Incident Response
- Reporting
- Continuous Learning

SIEM Solution

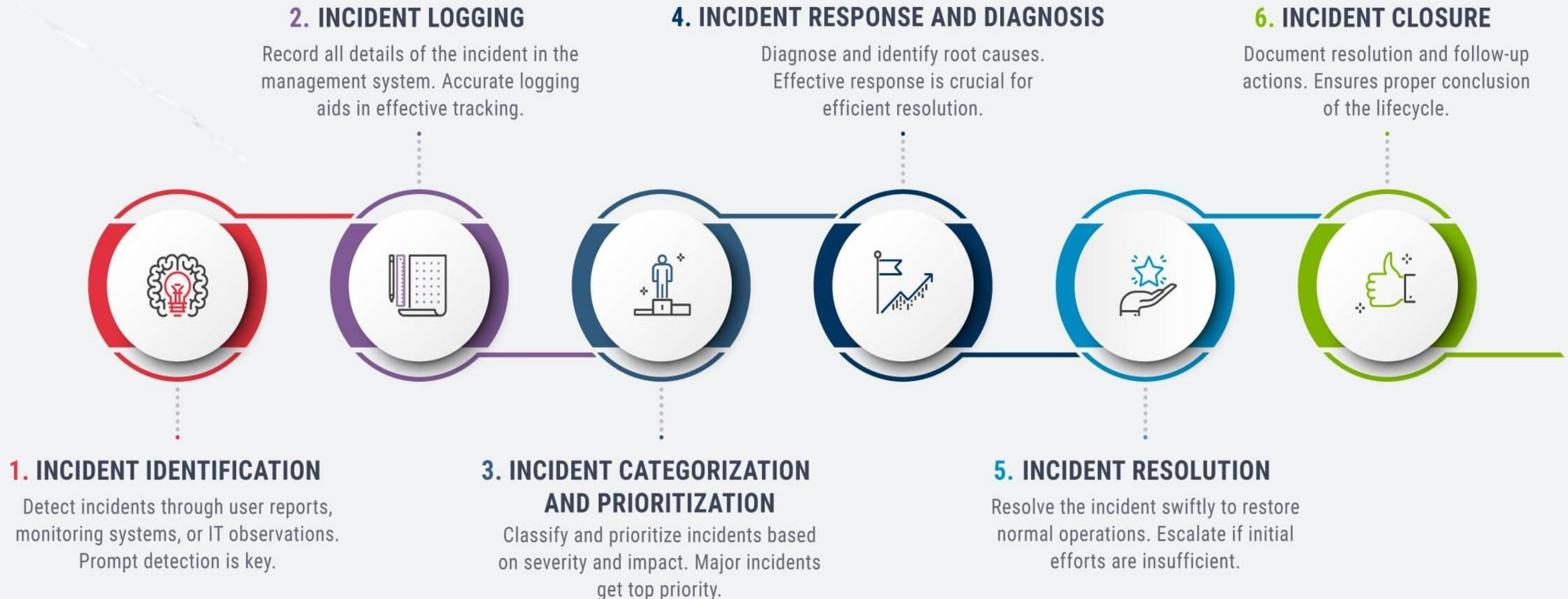


EDR Solution

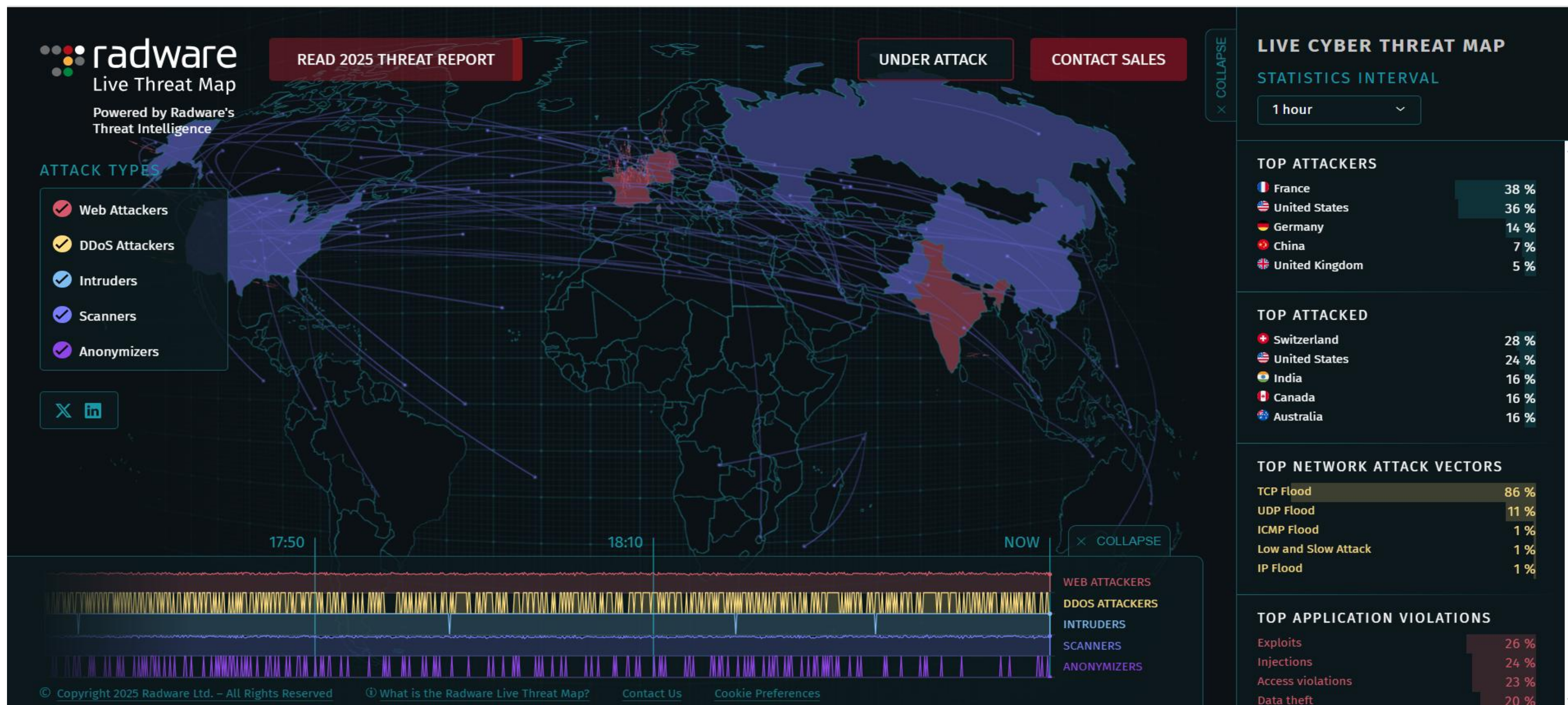


Incident Response

The Incident Management Lifecycle



Cyber threat Live Map



<https://livethreatmap.radware.com/>



Thank You

Arif Hussain

SOC Manager | Cybersecurity Professional

LinkedIn: [linkedin.com/in/arifhussainofficial](https://www.linkedin.com/in/arifhussainofficial)

Email: Arifhussain1@outlook.com